ASSIGNMENT

TITLE	Historic Cyber Attacks That Shaped Modern Cybersecurity
COURSE / PROGRAMME	Cybersecurity Fundamentals/Ethical Hacking
SUBMITTED BY	Raphael Ndidi-Uboma
SUBMITTED TO	Digital Eselsa
INSTITUTION / TRAINING PROVIDER	NIIT Enugu
DATE OF SUBMISSION	November 8, 2026

INTRODUCTION

Cybersecurity as we know it today was shaped by attacks that exposed weaknesses in computers, networks and human behaviour. This assignment explains six historic incidents, covering when each happened, how it worked, what damage it caused and what the world learned from it.

1. Creeper Virus (1971)

Year and creator: Creeper appeared in 1971. It was written by Bob Thomas, an engineer at BBN Technologies.

How it worked: Creeper was an experimental program that moved between computers on ARPANET, the early network that grew into the internet. It ran on DEC PDP-10 machines using the TENEX operating system. When it reached a computer, it displayed the message "I'm the creeper: catch me if you can!" and then jumped to another machine.

Who was affected and impact: Only a few research computers were affected. It was harmless and did not damage data or steal anything.

Significance: Creeper is widely seen as the first self-spreading program, an early ancestor of the computer worm.

Lessons learned: Ray Tomlinson later wrote a program called Reaper that travelled the network and deleted Creeper. Reaper is often called the first antivirus-style tool. The lesson was that programs can move between networked systems without a user's action, so defences must be designed for this.

2. Brain Virus (1986)

Year and creator: Brain appeared in 1986. It was created by two brothers, Basit Farooq Alvi and Amjad Farooq Alvi, in Lahore, Pakistan.

How it worked: Brain was a boot sector virus. It copied itself into the first sector of floppy disks used by IBM-compatible PCs. When someone started a computer using an infected disk, the virus loaded into memory and infected other disks that were inserted afterwards. It also changed the disk label to "(c) Brain".

Who was affected and impact: The brothers said they wanted to discourage illegal copying of their medical software. However, the virus spread worldwide through shared disks. It slowed

disk access and used up storage space, but it caused little permanent damage.

Significance: Brain is regarded as the first virus for the IBM PC.

Lessons learned: It showed that removable media can carry malware and that software piracy and security are linked. This led to the growth of antivirus software and the habit of scanning disks before use.

3. **Morris Worm (1988)**

Year and creator: The Morris Worm was released on 2 November 1988 by Robert Tappan Morris, a graduate student at Cornell University.

How it worked: The worm took advantage of weaknesses in Unix systems, including a flaw in the *fingerd* service, a debug feature in the *sendmail* email program, and weak passwords. It copied itself from computer to computer. A coding mistake made it infect the same machine many times, so computers became overloaded and crashed.

Who was affected and impact: About 6,000 computers, roughly 10 percent of the internet at that time, were slowed or disabled, including those at universities, research centres and military sites. Cleanup costs were estimated in the hundreds of thousands of dollars or more.

Significance: It was the first major internet worm. Morris was the first person convicted under the US Computer Fraud and Abuse Act.

Lessons learned: The attack led to the creation of the first Computer Emergency Response Team (CERT) at Carnegie Mellon University. It taught the importance of patching software, using strong passwords and coordinating incident response.

4. **ILOVEYOU (2000)**

Year and creator: ILOVEYOU spread in May 2000. It is linked to Onel de Guzman, a student in the Philippines.

How it worked: Victims received an email with the subject "ILOVEYOU" and an attachment named LOVE-LETTER-FOR-YOU.TXT.vbs. Windows often hid the real ".vbs" ending, so the file looked like harmless text. Opening it ran a script that overwrote many files and emailed itself to every contact in the user's Microsoft Outlook address book.

Who was affected and impact: Tens of millions of computers worldwide were infected, including those of companies, banks and government bodies. Damage was estimated at billions

of dollars, mainly from lost work and cleanup.

Significance: It showed how powerful social engineering is, because people open messages that seem to come from friends.

Lessons learned: The Philippines had no law covering the crime, so charges were dropped, and the country soon passed a law on electronic commerce and cybercrime. Companies began filtering risky attachments, showing full file extensions and training staff not to open unexpected files.

5. **Anna Kournikova Virus (2001)**

Year and creator: The virus appeared in February 2001. It was created by Jan de Wit, a young man from the Netherlands.

How it worked: It arrived as an email saying "Here you are" with an attachment called AnnaKournikova.jpg.vbs, which pretended to be a picture of the famous tennis player. Opening it ran a script that sent the same message to everyone in the victim's Outlook address book. De Wit built it using a ready-made "virus creation kit", so he needed very little skill.

Who was affected and impact: Email users worldwide were affected. It did not destroy files, but the flood of emails overloaded mail servers and slowed company networks.

Significance: It proved that beginners could create effective malware with downloadable tools. It also showed how celebrity names attract clicks.

Lessons learned: De Wit was sentenced to community service. Organisations learned to block script attachments, update email filters and teach users to be careful with attachments, even from known senders.

6. **Equifax Data Breach (2017)**

Year and creator: The breach took place between May and July 2017 and was announced in September 2017. In 2020, the US Department of Justice charged four members of China's military with carrying it out.

How it worked: Attackers used a known flaw in Apache Struts, a web application tool. A fix had been released months earlier, but Equifax had not installed it. The attackers entered the system and stayed hidden for weeks while collecting data.

Who was affected and impact: About 147 million people had personal data exposed, including

names, birth dates, addresses and Social Security numbers. Equifax agreed to a settlement of up to 700 million dollars.

Significance: It showed that even large companies holding sensitive data can fail at basic security.

Lessons learned: Organisations must patch quickly, keep an inventory of their systems, monitor networks, separate sensitive data and encrypt it. Leaders are now held more responsible for security failures.

7. **WannaCry Ransomware (2017)**

WannaCry spread in May 2017, locking files on more than 200,000 computers in over 150 countries and demanding Bitcoin payment. It used a Windows flaw called EternalBlue, even though Microsoft had released a patch earlier. It seriously disrupted the UK National Health Service, delaying appointments and surgeries. I think it had a major impact because it showed that ransomware can spread by itself and threaten human safety. It also pushed organisations to update systems promptly, keep offline backups and replace outdated software.

CONCLUSION

From Creeper's harmless experiment to Equifax's costly breach, each incident exposed a weakness: unpatched software, trusting users, weak design or poor planning. Together they led to antivirus tools, incident response teams, cybercrime laws and stronger security habits.

REFERENCES

1. Federal Bureau of Investigation (2020). Indictment of four Chinese military members for the Equifax hack. justice.gov.
2. US Federal Trade Commission (2019). Equifax data breach settlement. ftc.gov.
3. Eichin, M. & Rochlis, J. (1989). *With Microscope and Tweezers: The Worm from MIT's Perspective*.
4. Kaspersky. *A Brief History of Computer Viruses.* kaspersky.com.
5. Cybersecurity and Infrastructure Security Agency (CISA). Alert on WannaCry ransomware. cisa.gov.
6. Britannica. Entries on the Morris Worm and computer viruses. britannica.com.